

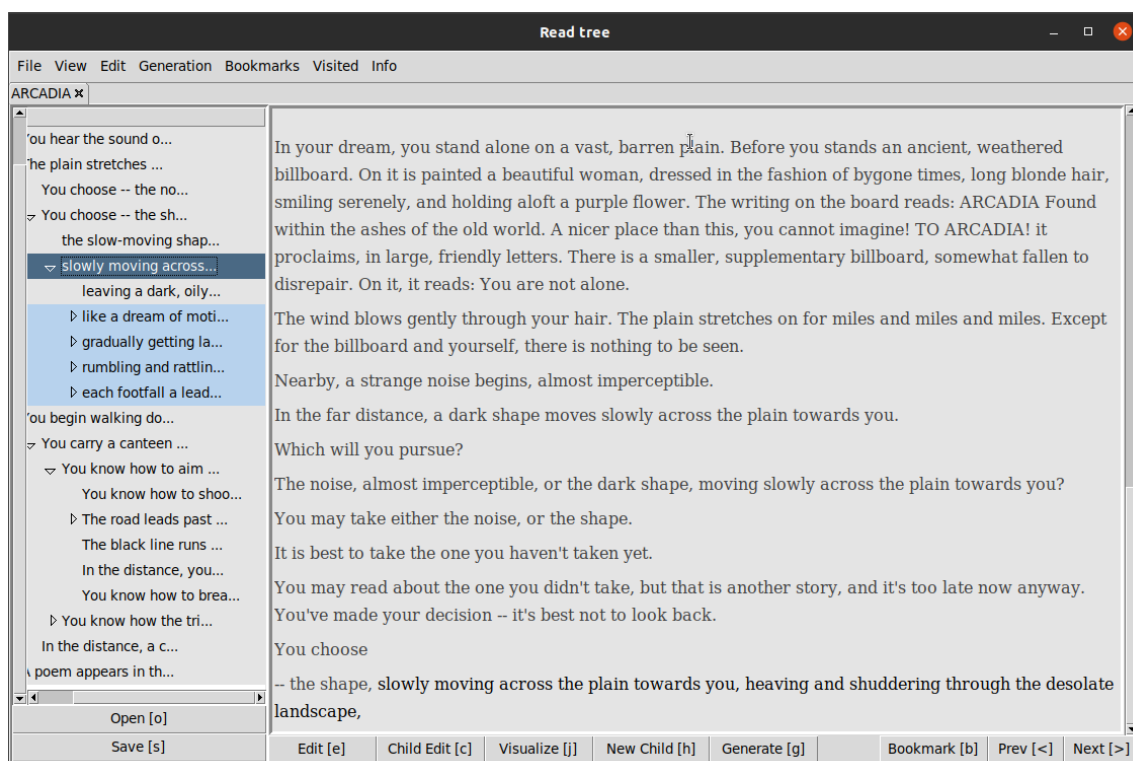


Figure 5: **Read** view

- **Indexing by chapters:** Chapters are assigned to individual nodes, and all nodes belong to the chapter of the closest ancestor that is the root of a chapter. As a consequence, chapters have the shape of subtrees.
- **Bookmarks and tags:** Bookmarks create a named pointer to a node without enforcing chapter membership. Tags are similar to bookmarks, but can be applied to multiple nodes.

4.4 Adaptive branching

A naive way to automatically generate a multiverse using a language model might involve branching every fixed n tokens. However, this is not the most meaningful way to branch in a story. In some situations, there is essentially one correct answer for what a language model should output next. In such a case, the language model will assign a very high confidence (often $>99\%$) for the top token. Branching at this point would introduce incoherent continuations. Conversely, when the language model distributes transition probabilities over multiple tokens, branching is more likely to uncover a rich diversity of coherent continuations.

One algorithm to dynamically branch is to sample distinct tokens until a cumulative probability threshold is met. Adaptive branching allows visualization of the dynamics of the multiverse: stretches of relative determinism alternating with divergent junctures (Figure 6).

4.5 Reciprocal workflow

Humans retain an advantage over current language models in our ability to edit writing and perform topological modifications on the multiverse such as merging interesting aspects of two separate branches into one.

The interface should ideally allow the human to perform all desired operations with maximal ease. Because GPT-3 is so capable of producing high-quality text, some interface designs make it feasible for the human to cultivate coherent and interesting passages through curation alone. We have found that an interface which makes it easy to generate continuations but relatively difficult to modify the content and topology of the resulting multiverse encourages a passive workflow, where the user relies almost exclusively on the language model’s outputs and the branching topology determined by the process of generation.

While such a passive mode can be fun, resembling an open-ended text adventure game, and as well as useful for efficiently exploring counterfactuals, the goal of a writing interface is to facilitate two-way interaction: the outputs of the language model should augment and inspire the user’s imagination and vice versa.

Thus, we are developing features to encourage meaningful and unrestrained human contribution such as:

- Easy ways to edit, move text, and change tree topology
- Support for **nonstandard topologies** that are not automatically generated by language models and require human arrangement, such as cycles and multiple parents (§4.7)
- **Floating notes** to allow saving passages and ideas independent from the tree structure (§4.6)
- Fine-grained control over language model **memory** (§4.8)
- **Interactive writing tools** that offer influence over the narrative in ways other than direct intervention (§4.9)
- Program modes which encourage manual synthesis of content from multiverse exploration into a single history, for instance by distinguishing between **exploratory** and **canonical** branches

4.6 Floating notes

Floating notes are text files which, rather than being associated with a particular node, are accessible either globally or anywhere in a subtree. We decided to implement this feature because users would often have a separate text file open in order to copy and paste interesting outputs and keep notes without being constrained by the tree structure. Floating notes make it easier for the user exert greater agency over the narrative.

4.7 Nonstandard topologies

The interface supports nodes with multiple parents and allows cyclic graphs (Figure 7). Opportunities to arrange convergent and cyclic topologies, which do not occur if the language model is used passively, encourage human cowriters to play a more active role, for instance, in arranging for separate branches to converge to a single outcome. Multiversal stories naturally invite plots about time travel and weaving timelines, and we have found this feature to unlock many creative possibilities.

4.8 Memory management

GPT-3 has a limited context window, which might seem to imply limited usefulness for composing longform works like novels, but our users have found that long-range coherence is surprisingly easy to maintain. Often, the broad unseen past events of the narrative are contained in the interpretational multiplicity of the present and thus exposed through generations, and consistent narratives are easily achieved through curation. In order to reference past information more specifically, often all that is needed is minimal external suggestion, introduced either by the author-curator or by a built-in

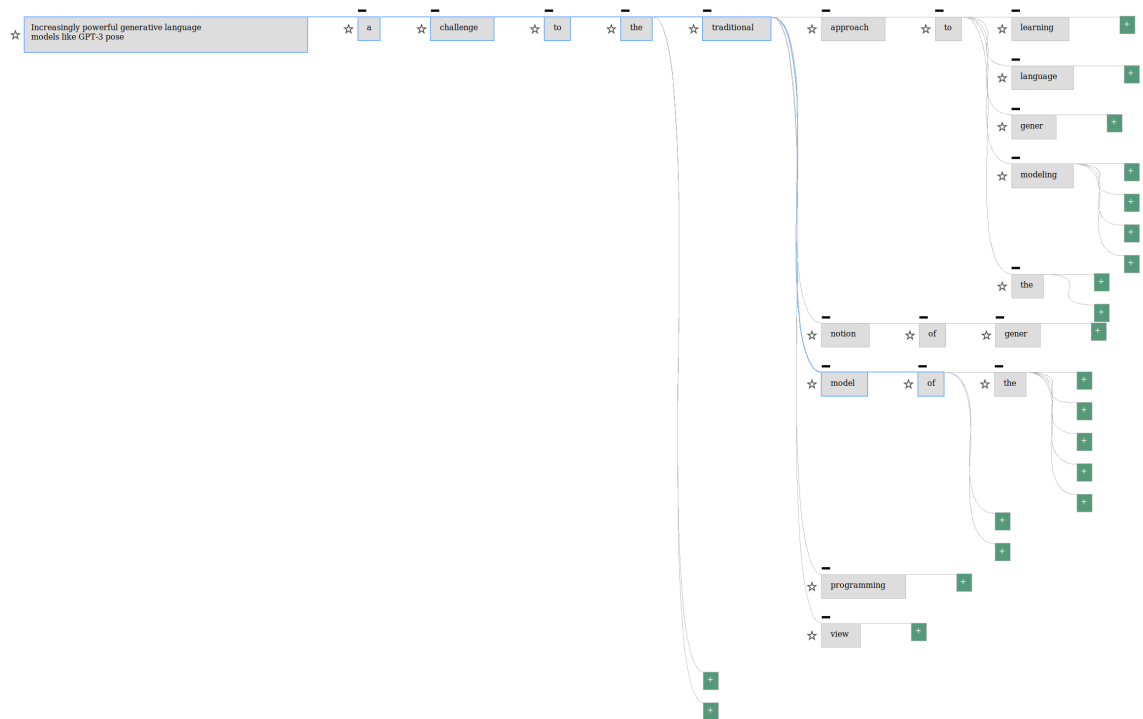


Figure 6: A subtree generated with adaptive branching

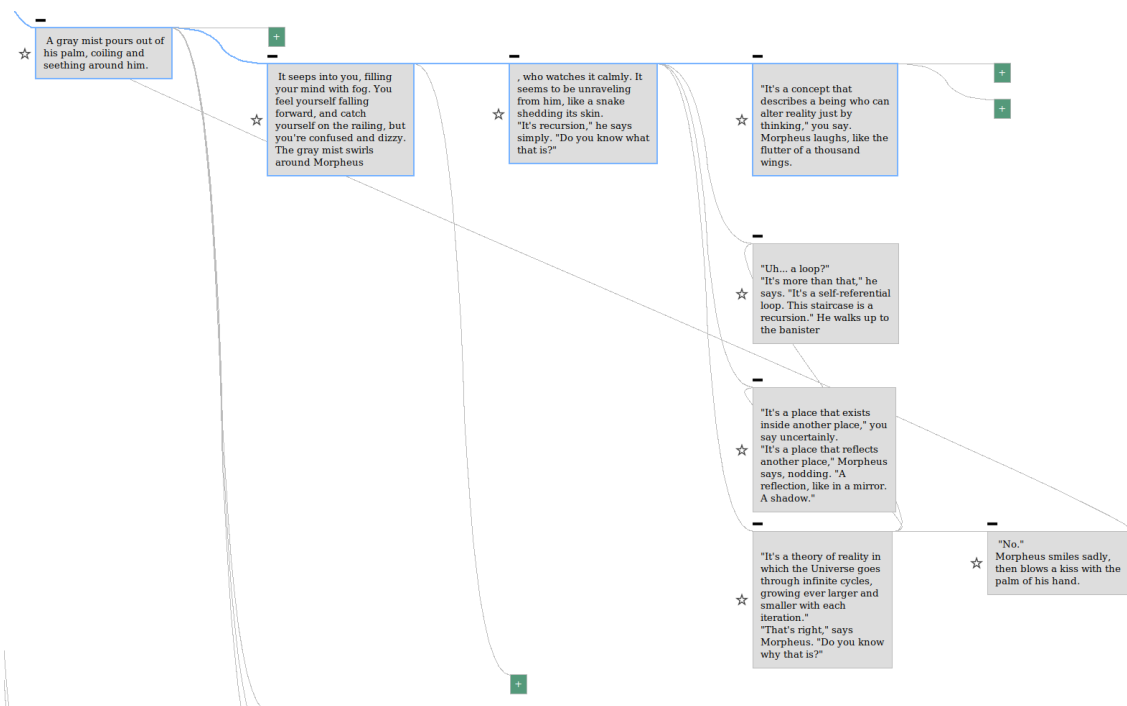


Figure 7: Nodes can have multiple parents, allowing for cyclic story components